DNS Guide: Wireshark Analysis of DNS Traffic

What is DNS?

DNS (Domain Name System) is the "phonebook" of the internet. It translates human-readable domain names (e.g., google.com) into machine-readable IP addresses (e.g., 142.250.190.78).

DNS operates primarily over **UDP port 53**, but can fall back to **TCP port 53** for large responses (zone transfers, DNSSEC, etc.).

Typical Ports

- **UDP 53** — Standard DNS queries and responses (most common)
- **TCP 53** — Large responses, zone transfers (AXFR), DNS over TLS (DoT)

Why Analysts Care About DNS

- Almost every connection starts with a DNS lookup.
- Attackers heavily abuse DNS for **Command & Control (C2)**, data exfiltration, and domain generation algorithms (DGAs).
- DNS can leak sensitive information (internal hostnames, software versions).
- Malicious domains, phishing sites, and malware callbacks are often discovered through DNS analysis.

Wireshark Fundamentals for DNS Analysis

1. Basic Filtering

Bash

```bash
dns                          # All DNS traffic
dns.flags.response == 0      # DNS Queries only
dns.flags.response == 1      # DNS Responses only
dns.qry.name contains "google"   # Search for specific domain
dns.qry.type == 1            # A records (IPv4)
dns.qry.type == 28           # AAAA records (IPv6)
dns.qry.type == 5            # CNAME
dns.qry.type == 16           # TXT records (often used for exfil)
dns.flags.rcode != 0         # Failed responses (NXDomain, etc.)
```

2. Key DNS Fields to Examine

Query Section

- Queries → Domain name being asked (qry.name)
- Query Type (qry.type): A, AAAA, MX, NS, TXT, CNAME, PTR, etc.
- Query Class (usually IN for Internet)

Response Section

- Answers — The resolved records
- Authoritative answers vs Non-authoritative
- Time To Live (TTL) — How long the record can be cached
- Additional records (sometimes contains extra info)

Flags

- Recursion Desired (RD) — Client asks server to resolve recursively
- Recursion Available (RA) — Server supports recursion
- Response Code (rcode): 0 = No error, 3 = NXDomain (does not exist)

Common DNS Analysis Scenarios in Thaghrah Challenges

1. **Basic Resolution**
    - Find what IP a domain resolved to.
2. **TXT Record Exfiltration**
    - Attackers hide data in TXT records (very common in real malware).
3. **CNAME Chaining**
    - Multiple CNAMEs leading to the final A record.
4. **Encoded or Chunked Data**
    - Data split across multiple queries or encoded in subdomains (e.g. data1.example.com, data2.example.com).
5. **Malicious Domain Detection**
    - Suspicious domain names (random strings, unusual TLDs).
    - High volume of DNS queries (possible DGA malware).
6. **DNS Tunneling**
    - Very long subdomains or unusually large DNS packets.

Practical Workflow in Wireshark

1. Open the PCAP and apply dns filter.
2. Expand the **Domain Name System (query)** or **(response)** section.
3. Use **Follow → UDP Stream** (or TCP if applicable) to see the full conversation.
4. Go to **Statistics → DNS** for:
    - Domain statistics
    - Query/Response breakdown
5. Export DNS packets or use **tshark** for automation if needed.

Pro Tips:

- Right-click a DNS response → **Copy** → **Value** on the answer field to quickly grab IPs or text.
- Use dns contains "flag" or dns contains "THAGHRAH" to hunt for hidden flags.

Security Red Flags in DNS Traffic

- TXT records containing base64-looking strings.

- Extremely long domain names (DNS tunneling).
- Queries for suspicious or newly registered domains.
- High number of NXDomain responses (DGA behavior).
- DNS responses containing internal/private IP addresses.
- Zone transfer attempts (AXFR queries).

Example Challenge Goals You Will Encounter

- Extract the hidden flag from a TXT record.
- Reconstruct a message sent via multiple encoded DNS queries.
- Identify the real destination IP a user was trying to reach through CNAME chaining.
- Spot signs of DNS tunneling or data exfiltration.
- Find the command-and-control domain used by "malware" in the capture.

Tips for Thaghrah Students:

- DNS traffic is usually very clean and easy to read compared to TCP or TLS.
- Always check both **Queries** and **Answers** sections.
- Look for unusual record types (especially TXT and CNAME).
- Combine with other protocols: e.g., dns || http to see the full picture.